



Planning Development: SyIAR Gemilang

(Sistem Informasi Al-Azhar Rumah Gemilang)



1. Overview Proyek

Item	Deskripsi
Nama Aplikasi	SyIAR Gemilang
Arsitektur	Decoupled Full-Stack: CI4 sebagai REST API Backend + Express.js sebagai SSR Frontend
Fokus Fase 1	1. Sistem Role & Permission Dinamis (RBAC) 2. Modul Penilaian Siswa/Santri
Target Pengguna	Super Admin, Admin Akademik, Guru/Asatidz, Kepala Bagian
Lingkungan	Node.js 18+/20+, PHP 8.1+, MySQL 8.x, npm/composer



2. Arsitektur & Tech Stack

Layer	Teknologi	Peran
Frontend Server	Express.js + EJS/Handlebars	Routing, SSR, UI Rendering, Session Management, API Consumer
Styling	Tailwind CSS (via PostCSS)	Utility-first CSS, custom components, responsive design
Backend API	CodeIgniter 4 (RESTful)	Business Logic, Database Access, Validation, JWT/Session Auth, JSON Response
Database	MySQL 8.x / MariaDB	Relational data storage, ACID transactions
HTTP Client	Axios / Node fetch	Komunikasi Express → CI4
State/Cache	Redis / Memory Cache (CI4)	Caching permission mapping & session token
Build Tool	Vite / npm scripts	Compile Tailwind, optimize assets, watch

 Alur Data:



3. Desain Database (MySQL)

(Skema tetap fokus pada RBAC & Penilaian. CI4 akan mengakses tabel ini langsung.)

Tabel	Fungsi Utama
users	Akun pengguna, password hash, status
roles	Master role (admin, guru, wali, dll)
permissions	Hak akses granular (modul.create , modul.read , dll)
role_permissions	Mapping many-to-many role ↔ permission
user_roles	Mapping many-to-many user ↔ role
santri	Data master siswa (NIS, kelas, status)
kategori_penilaian	Kategori besar (Akhlak, Tahfidz, Akademik)
aspek_penilaian	Sub-aspek dinamis + skala min/max + bobot
penilaian	Transaksi nilai per santri, aspek, pengajar, tanggal
log_aktivitas	Audit trail (login, ubah nilai, hapus data)

 **Catatan:** Gunakan CI4 Migrations untuk deployment skema. Aktifkan `softDeletes` pada `users` & `santri`. Index pada `NIS`, `role_id`, `santri_id`, `aspek_id`.

4. Sistem Role & Permission (RBAC Dinamis)

4.1. Konsep

- Role & Permission **100% dinamis** via UI Admin.
- Format permission: `modul.aksi` (contoh: `penilaian.create`, `master.santri.export`).

- Satu user boleh memiliki **multi-role** (union permission).
- Validasi dilakukan di **2 layer**:
 - i. **Express.js**: Middleware route guard (menyembunyikan menu/route sebelum render)
 - ii. **CI4 API**: Filter/API Guard (memvalidasi token & permission sebelum eksekusi query)

4.2. Struktur Permission Table

```
permissions
  id (PK) | kode (VARCHAR, UNIQUE) | modul (VARCHAR) | aksi (VARCHAR) | deskripsi
```

4.3. Alur Autentikasi & Otorisasi

1. User login di Express → Express hit `POST /api/auth/login` ke CI4.
2. CI4 validasi → return `{ token: "JWT/SessionID", user: {...}, permissions: [...] }`
3. Express simpan `token` di `httpOnly cookie`, simpan `permissions` di session.
4. Express middleware `requirePermission('penilaian.create')` cek session sebelum render route.
5. Setiap request Express → CI4 menyertakan `Authorization: Bearer <token>` di header.
6. CI4 `AuthFilter` decode token & cek `hasPermission()` di DB/cache.

5. Modul Penilaian Siswa/Santri

5.1. Fitur UI (Express + Tailwind)

Fitur	Komponen Tailwind/JS
Dashboard Guru	Card grid, Chart.js (progres nilai), quick-access button
Input Penilaian	Form dinamis, <code>input[type=range]</code> / <code>number</code> , live validation, draft auto-save
Rekap & Export	DataTables + Tailwind table, export CSV/Excel via CI4 endpoint
Manajemen Aspek	Modal CRUD, drag-drop urutan (opsional), toggle aktif/nonaktif

5.2. API Endpoint CI4 (Contoh)

Method	Endpoint	Deskripsi	Auth Required
GET	/api/santri	List santri (filter kelas/status)	✓
GET	/api/penilaian/aspek/:kategori_id	Fetch aspek aktif	✓
POST	/api/penilaian/submit	Simpan nilai batch	✓ penilaian.create
GET	/api/penilaian/rekap?periode=2024-2	Rekap nilai + bobot	✓ penilaian.read
GET	/api/laporan/export/excel	Download file	✓ penilaian.export

6. Integrasi Express.js + Tailwind CSS

6.1. Struktur Folder Frontend

```
frontend/  
├── src/  
│   ├── views/           # EJS templates (layout, pages, partials)  
│   ├── public/          # Static assets (compiled CSS, images, JS)  
│   ├── routes/          # Express routers (auth, dashboard, penilaian, admin)  
│   ├── middleware/      # auth.js, permission.js, error.js  
│   ├── services/        # apiClient.js (Axios instance with token injection)  
│   └── utils/           # formatter.js, validator.js  
├── tailwind.config.js  
├── postcss.config.js  
├── package.json  
└── server.js
```

6.2. Setup Tailwind di Express

```
# 1. Inisialisasi  
cd frontend && npm init -y  
npm install express ejs axios dotenv  
npm install -D tailwindcss postcss autoprefixer concurrently  
  
# 2. Init Tailwind  
npx tailwindcss init -p
```

tailwind.config.js

```
module.exports = {
  content: [ './src/**/*.ejs,js', './public/**/*.js,css' ],
  theme: { extend: { colors: { primary: '#0f766e', secondary: '#f59e0b' } } },
  plugins: [],
}
```

postcss.config.js

```
module.exports = {
  plugins: [ require('tailwindcss'), require('autoprefixer') ],
}
```

package.json scripts

```
"scripts": {
  "build:css": "tailwindcss -i ./src/styles.css -o ./public/css/main.css --minify",
  "dev:css": "tailwindcss -i ./src/styles.css -o ./public/css/main.css --watch",
  "start": "node server.js",
  "dev": "concurrently \"nodemon server.js\" \"npm run dev:css\""
}
```

7. Alur Komunikasi & Security Flow

Layer	Mekanisme Keamanan
Express → Client	<code>res.cookie('syiar_token', token, { httpOnly: true, secure: true, sameSite: 'strict' })</code>
Express → CI4	Axios interceptor attach headers: { Authorization: Bearer \${token} }
CI4 Auth	<code>Config/Filters.php</code> → <code>AuthFilter</code> validasi JWT → <code>PermissionFilter</code> cek DB/Cache
CI4 → DB	Query Builder + Prepared Statements, <code>strict mode</code> aktif
CORS	<code>Config/CORS.php</code> di CI4: <code>allowedOrigins: ['http://localhost:3000']</code> , <code>allowedHeaders: ['Authorization', 'Content-Type']</code>

17 8. Rencana Pengembangan (Phases & Timeline)

Fase	Durasi	Deliverable	Target
Phase 1	Minggu 1	- Setup CI4 API + CORS + JWT - Setup Express + EJS + Tailwind build - DB Migration awal	Infrastruktur siap, login f bw working
Phase 2	Minggu 2	- CRUD Role & Permission (UI + API) - Middleware <code>checkPermission</code> di Express & CI4 - Menu dinamis berdasarkan role	RBAC dinamis 100% fungsional
Phase 3	Minggu 3	- CRUD Master Santri + Import CSV - CRUD Kategori & Aspek Penilaian	Data master siap, validasi skala aktif
Phase 4	Minggu 4	- Form Input Penilaian (batch + draft) - Rekap, Chart progres, Export Excel	Modul penilaian siap produksi
Phase 5	Minggu 5-6	- UI Polish (Tailwind components, loading states) - Log aktivitas, error handling global - Testing, dokumentasi, deployment	Release v1.0 stable

9. Keamanan & Best Practices

Aspek	Implementasi
Auth Token	JWT (exp 2h) + Refresh Token, disimpan di <code>httponly</code> cookie
CSRF	Express <code>csrf</code> middleware + CI4 CSRF filter untuk form submit
Rate Limiting	<code>express-rate-limit</code> pada endpoint <code>/api/auth/*</code>
Input Sanitization	<code>validator.js</code> di Express, CI4 <code>validation</code> library di backend
-	Global <code>try/catch</code> , custom error response JSON, log ke

Error Handling	writable/logs/
Backup	mysqldump cron + Express backup trigger via admin UI
Environment	.env terpisah: frontend/.env , backend/.env (JANGAN commit!)

10. Langkah Setup Awal (Quick Start)

10.1. Backend (CodeIgniter 4)

```
composer create-project codeigniter4/appstarter backend
cd backend
cp env .env
# Edit .env: CI_ENVIRONMENT = development, database credentials
php spark serve --port 8080
```

Aktifkan CORS di `config/CORS.php` & buat controller API di `app/Controllers/Api/`.

10.2. Frontend (Express + Tailwind)

```
mkdir frontend && cd frontend
# Ikuti panduan setup Tailwind di Section 6.2
npm install
npm run dev # Mulai Express (port 3000) + Tailwind watch
```

10.3. Integrasi Awal

- Express buat route `/login` → POST ke `http://localhost:8080/api/auth/login`
- Simpan token → Redirect ke `/dashboard`
- Dashboard fetch `http://localhost:8080/api/user/permissions` → Render menu dinamis

Lampiran & Fase Lanjutan

- Portal Wali Santri (view-only via token scoped)
- Presensi QR/RFID integration
- Export Rapor PDF (dompdf/mpdf via CI4)
- Notif kasi WhatsApp Gateway (Fonnte/Wablas API)
- CI/CD Pipeline (GitHub Actions → VPS/Hosting)



Dokumen ini bersifat living document. Update saat requirement berubah atau hasil review testing.

Arsitektur: Express.js (SSR + Tailwind) ↔ CI4 REST API ↔ MySQL

Versi: 1.1.0 | **Status:** Planning Approved | **Tanggal:** 2024

💡 Catatan Penting untuk Implementasi:

- **Token Flow**:** Gunakan JWT yang di-generate CI4, lalu Express menyimpannya sebagai session.
- **Tailwind Build**:** Jangan gunakan CDN Tailwind untuk produksi. Gunakan build script yang disediakan.
- **CI4 sebagai API**:** Matikan CSRF untuk endpoint `/api/\*`, atau gunakan token-based authentication.
- **Middleware Permission**:** Buat helper di Express:

```
```js
// middleware/permission.js
module.exports = (perm) => (req, res, next) => {
 const userPerms = req.session?.permissions || [];
 if (userPerms.includes(perm)) return next();
 req.flash('error', 'Akses ditolak');
 res.redirect('/dashboard');
};
```